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A Comprehensive Evaluation of GPM IMERG Satellite Rainfall Data Across Multiple Temporal and Spatial Scales for Sustainable Flood Risk Management in East Java, Indonesia

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Abstract

Accurate extreme rainfall representation is critical for resilient hydrological design and sustainable water management in tropical regions. This study evaluates the GPM IMERG product across three diverse watersheds in East Java (Welang, Kedak, and Grindulu) using Extreme Value Theory (EVT). By employing Generalized Extreme Value (GEV) and Peaks Over Threshold (POT) methods, the research assesses the reliability of satellite estimates in characterizing the extreme events that safeguard community security and infrastructure longevity. Results indicate that while GPM IMERG excels at monthly scales, it lacks the daily precision required for effective flash flood mitigation, particularly in small basins. Crucially, GEV analysis reveals a structural mismatch: ground observations exhibit heavy-tailed (Fréchet) distributions, while GPM IMERG follows bounded (Weibull) distributions. Consequently, the satellite product underestimates high-magnitude events at long return periods, the exact events that define the design limits of adaptive hydraulic structures. Complementary POT analysis identifies scale-dependent biases across catchments. These findings suggest that while GPM IMERG is robust for regional monitoring, it requires distribution-specific bias correction to support disaster-resilient engineering. Addressing these gaps is essential for achieving climate-responsive sustainable development in data-scarce regions.

Keywords: GPM IMERG; GEV; climate resilience; sustainable infrastructure; POT; East Java

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1. Introduction

Accurate estimation of extreme rainfall and long-term return periods is fundamental to resilient water resources engineering, particularly for sustainable flood risk mitigation in regions prone to catastrophic hydrometeorological events. While traditional hydrological analyses in Indonesia rely heavily on ground-based rain gauge networks, the inherent spatial scarcity and uneven distribution of these stations, especially across the complex topographies of East Java, often fail to capture the localized intensity of extreme storms [1,2]. This limitation introduces significant uncertainties in design discharge estimation, where even minor data gaps can lead to the under-design of critical infrastructure.

As an alternative, satellite-based precipitation products, particularly those from the Global Precipitation Measurement (GPM) mission, have significantly complemented ground observations in the last two decades [3]. Their Integrated Multi-Satellite Retrievals (IMERG) product provides high spatial and temporal resolution rainfall estimates, enabling

coverage of regions with sparse ground monitoring networks [4]. However, GPM retrieval performance varies across local spatio-temporal scales, necessitating rigorous evaluation. In practical hydrological analysis, numerous studies have increasingly used satellite rainfall data obtained through NASA's Giovanni platforms. These included GPM IMERG and TRMM Multi-satellite Precipitation Analysis (TMPA) [5,6], with the essence of supporting rainfall assessment, calibration, and validation in data-limited regions.

Globally, the evaluation of GPM IMERG products has evolved through a range of statistical and hydrological approaches [7,8]. Previous results showed that a comprehensive evaluation was required to effectively characterize the uncertainties of satellite-based precipitation products across different regions, temporal scales, surface features, and rainfall regimes. The TMPA and IMERG products publicized by NASA Giovanni have also been widely examined. In addition, it was reported that GPM IMERG had been extensively assessed across various global regions [9–12]. Several studies consistently reported that IMERG provided broader spatial coverage, finer spatial–temporal resolution, and generally more accurate rainfall estimates compared to the earlier TMPA products [9]. Some initial assessments conducted in Brazil [13], China [14–16], Pakistan [17], United States [18], the upper Nile basin of Egypt [19], the Netherlands [20], and Singapore [11] found that the product was suitably used in hydro-climatic applications.

Based on this perspective, several studies on the use of satellite-based rainfall data have been conducted in Indonesia. The reliability of TMPA satellite rainfall data was examined across three sub-basins of the Brantas Watershed [2]. The results showed that TMPA data achieved ideal monthly performance (correlation ≈ 0.8) and produced area-averaged extreme rainfall estimates with errors less than 10%, although daily rainfall accuracy was relatively low. As an advancement, this research studied whether GPM exhibited better performance than TMPA data. Similarly, ref. [21] reported that GPM IMERG accuracy improved alongside temporal aggregation, with monthly and annual rainfall showing correlations greater than 0.7. The daily accuracy was highly variable, and the hourly performance was perceived as poor. At shorter time scales, IMERG tended to overestimate overall rainfall, yet extreme brief-duration events were often underestimated due to spatial smoothing effects. Prior research on the comparison between GPM IMERG and TMPA products in the Sampean Baru, Bedadung, and Mayang watersheds reported that the GPM-IMERG showed slightly better results than the TRMM 3B42 [22]. However, these studies largely focused on standard statistical performance at daily or monthly resolutions, leaving a gap in understanding how these products behave during rare, extreme events.

This research aims to bridge the existing knowledge gap by evaluating the reliability of GPM IMERG precipitation estimates across three distinct sub-basins in East Java (Grindulu, Kedak, and Welang), which represent a diverse range of catchment scales and gauge densities [23]. Unlike previous analyses that focused on standard statistical metrics, this study explicitly examines the implications of IMERG-derived rainfall for return-period-based extreme rainfall estimation. The primary motivation for utilizing Extreme Value Theory (EVT) is to critically compare the performance of the Generalized Extreme Value (GEV) and Peaks Over Threshold (POT) methods, specifically focusing on their stability when dealing with limited data records. By analyzing the structural differences in distribution tails, alongside factors like rainfall intensity and gauge representation, this study provides a novel diagnostic tool for understanding the inherent discrepancies between satellite-derived estimates and ground observations during intense tropical convective systems. Ultimately, the integration of multi-temporal evaluation with tail-behavior analysis across multiple watershed scales offers a new and robust framework that can serve as a technical reference for hydrological infrastructure planning and flood risk assessment in data-scarce tropical regions.

2. Research Area, Datasets, and Methodology

2.1. Research Area

The research area consisted of three watersheds in East Java Province. These included the Grindulu (Pacitan), Kedak (Kediri), and Welang (Pasuruan), as shown in Figure 1. The watersheds were selected to represent a variation in basin size, ranging from small to large catchments, as well as differences in spatial distribution and density of rain gauge stations used for estimating area-averaged rainfall for calibration purposes. In addition, the catchments covered a wide range of drainage areas, enabling the evaluation of satellite-based rainfall performance across different spatial scales. This was essential considering that basin size significantly influenced rainfall spatial variability and satellite retrieval accuracy. The watersheds also varied in terms of respective ground gauge distribution and network density, which provided a suitable basis for assessing the relationship between gauge representativeness and GPM IMERG calibration performance. Several of the watersheds were categorized as critical catchments with relevance to regional flood management, water resources planning, and agricultural water supply, making them suitable pilot basins for evaluating satellite rainfall applicability in operational hydrological assessment.

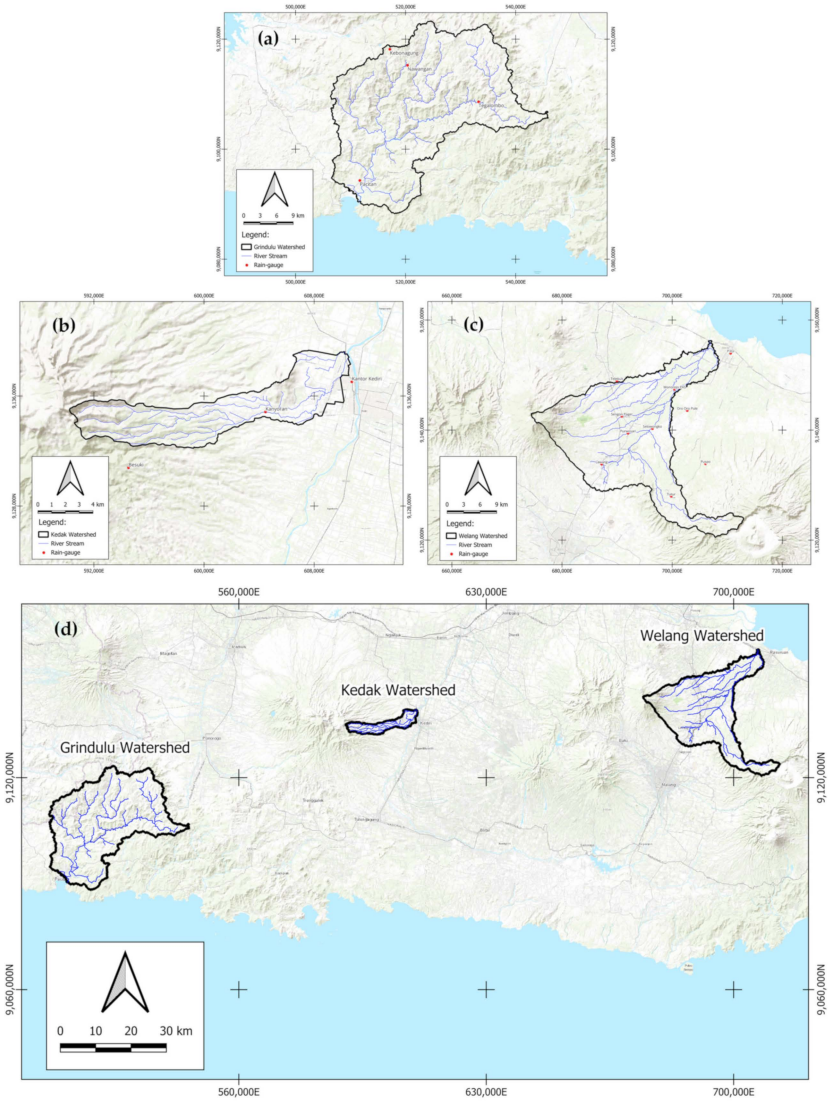


Figure 1. Research locations: (**a**) Grindulu watershed, (**b**) Kedak watershed, (**c**) Welang watershed, and (**d**) All watershed location.

The average basin slopes of the investigated watersheds exhibit a significant gradient of topographical complexity. The Grindulu watershed represents the lower bound of the dataset with a slope of 0.0183, indicating a relatively subdued relief compared to other watersheds in the study. In contrast, the Welang watershed demonstrates a moderate increase in topographical steepness with a recorded value of 0.0332. The most pronounced gradient is observed in the Kedak watershed, which features a maximum average basin slope of 0.06284. This value signifies a markedly higher degree of vertical relief, representing an approximately twofold increase relative to the Welang basin and more than triple the gradient of the Grindulu basin.

The rain gauge density varies significantly, showing a direct relationship between a smaller catchment area and a higher density of stations. For instance, the Kedak watershed (52 km^2) possesses the highest density at 14 km^2/station from its three monitoring sites. As the watersheds get larger, the coverage thins: the Welang watershed (511.6 km^2) has a density of 51 km^2/station across its 10 sites, while the Grindulu watershed (715 km^2) shows the sparsest coverage, with only 179 km^2/station provided by its four stations. Furthermore, detailed technical information and geographical characteristics of each watershed are shown in Table 1.

Table 1. Characteristics of the watershed research and distribution of rain gauge stations.

No	Watershed	City	Catchment Area (km^2)	Average Basin Slope	Rain Gauged Density (km^2/Station)	Rain Gauge Stations
1	Kedak	Kediri	52	0.0628 (63–1540) m a.s.l.	14	3 (Kediri, Besuki, Kanyoran)
2	Welang	Malang-Pasuruan	511.6	0.0332 (5–1690) m a.s.l.	51	10 (Lawang, Purwosari, Sengon Pager, Telebuk, Selowongko, Wonorejo PAS, Oro-oro Pule, P3GI, Puspo, Tutur)
3	Grindulu	Pacitan	715	0.0183 (2–860) m a.s.l.	179	4 (Kebon Agung, Nawangan, Pacitan, Tegal ombo)

2.2. Datasets

2.2.1. GPM IMERG

The GPM IMERG is a unified satellite-based precipitation product developed under the GPM mission, led by NASA and JAXA [6,24–26]. IMERG algorithmically combined data from passive microwave sensors on a constellation of international satellites, infrared observations, and surface precipitation gauge measurements to generate continuous, near-global estimates of both rainfall and snowfall.

The main product of this system, the IMERG Final Run, provided high-resolution precipitation data at spatial and temporal resolutions of $0.1^\circ \times 0.1^\circ$ (approximately 10 km × 10 km) and 30 min, respectively [27]. The detailed level made it valuable for hydrological modeling and climate research, specifically in regions with sparse ground-based gauge networks [28]. The IMERG algorithm adopted advanced methods. These included morphing (the CMORPH algorithm) and Kalman filter-based statistical merging methods, used to integrate and calibrate the various satellite retrievals with ground reference data. As a result, the IMERG Version 07 (Final Run) dataset was used in the present research. This version benefited from improved calibration with gauge-based data from the Global Precipitation Climatology Centre (GPCC), of which IMERG's final run significantly enhanced the accuracy and reliability for long-term climate and water resources analyses, particularly in tropical regions, compared with both early and late runs for GPM IMERG data. Several studies reported that the IMERG-F had gauge correction at the monthly

scale, leading to the exceptional performance compared to both -E and -L run products as shown in China [16,29], East Asia [10], Austria (temperate climate) [30], Saudi Arabia [31], Pakistan [32], and Italy [33].

The GPM IMERG record is prominent for its extensive temporal coverage. Data recording started in June 1998 and extended to the present (2025), providing a continuous and consistent observational dataset covering over 27 years. This long-term record is invaluable for analyzing precipitation trends, variability, and extreme weather events associated with the changing climate, specifically for tropical regions such as Indonesia.

2.2.2. Rain Gauge Data

Ground-based rainfall observations were obtained from the respective river basin organizations responsible for the Kedak, Grindulu, and Welang watersheds. The available daily rainfall time series record length varied among stations, with a minimum observation period of 10 years. Detailed information on data availability and station distribution for each watershed is shown in Figure 2.

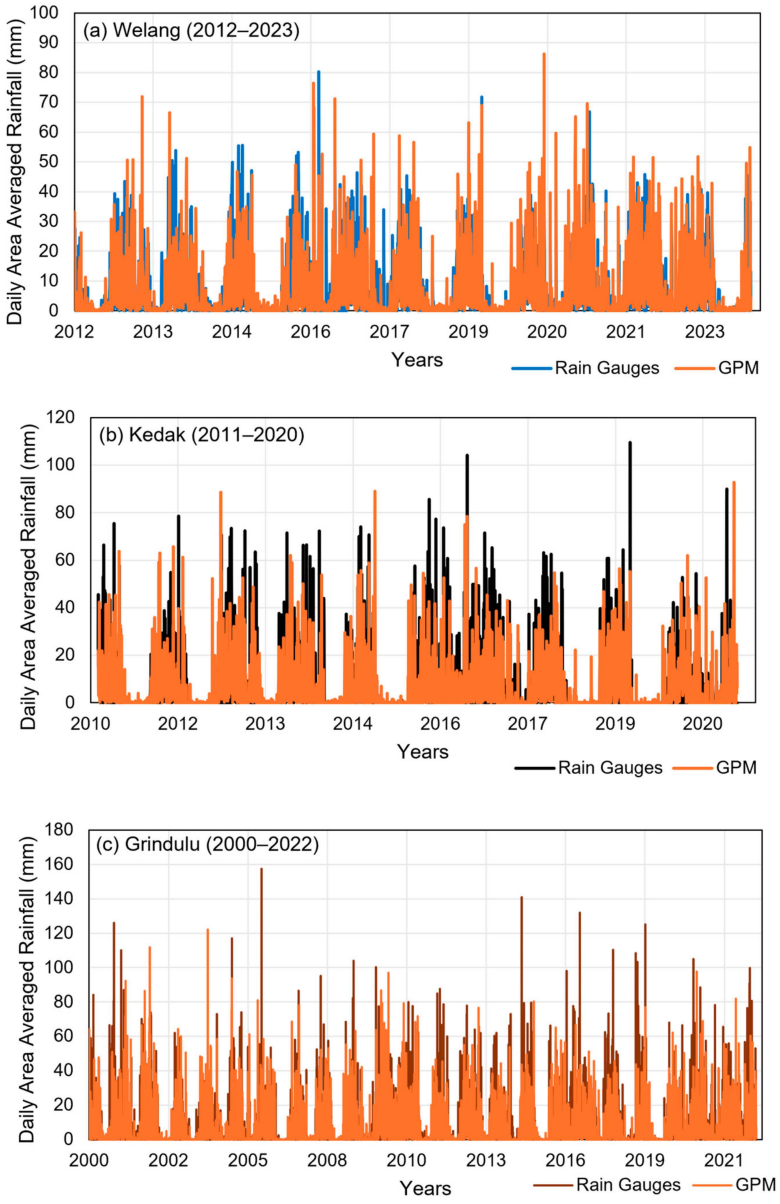


Figure 2. Daily area-averaged rainfall time series data of observed rain gauges and GPM data: (**a**) Welang watershed, (**b**) Kedak watershed, and (**c**) Grindulu watershed.

Daily rainfall measurements from all gauges located within each watershed were used to derive spatially representative areal rainfall. Station observations were spatially averaged using the Thiessen polygon method. The Thiessen polygon method was selected because of the relatively low density and uneven distribution of rain gauges in these specific catchments (e.g., only 3 gauges in Kedak and 4 in Grindulu). More complex spatial interpolation methods like Kriging require a higher density of data points to generate reliable semi-variograms. The essence was to account for the spatial influence of individual gauges daily. The resulting areal mean rainfall time series was subsequently used for comparison with the GPM IMERG satellite precipitation estimates. This spatial aggregation method minimized sampling bias arising from uneven rain gauge distribution and ensured consistency between point-based observations and gridded precipitation products. The entire process allowed a direct comparison based on equivalent area-averaged rainfall. To ensure temporal consistency, spatial averaging was performed daily by aggregating observations from all functional stations within the same day.

Prior to the spatial aggregation process, a rigorous Quality Control (QC) protocol was applied to the raw rain gauge data to ensure its reliability as the ground reference. First, to eliminate temporal bias and uncertainties, a strict completeness criterion was enforced: only stations possessing a continuous, uninterrupted daily data record (0% missing data) throughout the 10-year observation period were selected. Consequently, mathematical gap-filling or spatial interpolation was not required. Second, the stationarity and consistency of the time series were verified using Double Mass Curve (DMC) analysis. The resulting cumulative precipitation plots exhibited consistent linear relationships without significant inflection points, validating that the dataset is homogeneous and free from non-climatic artificial shifts. Following this QC validation, the verified station data were spatially averaged using the Thiessen polygon method. Consistency testing via Double Mass Curve (DMC) analysis (Figure 3) yielded strictly linear scatter plots (R^2 = 0.99) for all stations across the three watersheds, verifying the consistency of the dataset.

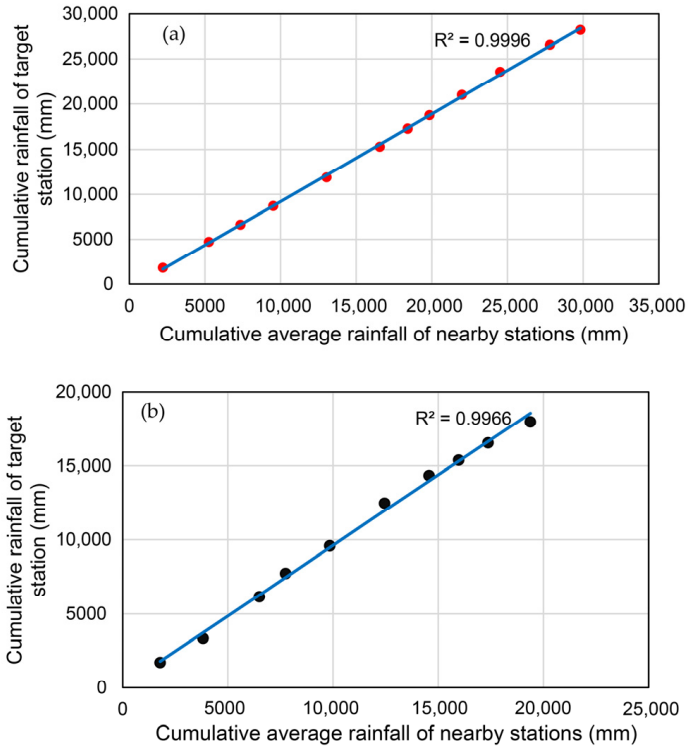


Figure 3. *Cont.*

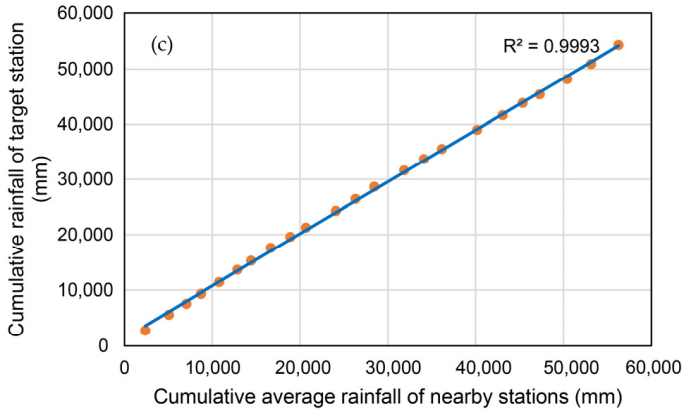


Figure 3. Double Mass Curve (DMC) plot for consistency testing of representative rain gauges: (**a**) Lawang station (Welang), (**b**) Kanyoran station (Kedak), and (**c**) Nawangan station (Grindulu).

2.3. Statistical Performance Evaluation Methodology

A suite of statistical metrics was adopted to quantitatively assess the accuracy and reliability of the GPM IMERG precipitation product against ground-based rain gauge observations. These metrics evaluated different aspects of agreement, from overall bias and correlation to the critical capture of extreme values. Furthermore, the analysis was conducted on both daily and monthly time scales to comprehensively understand the GPM IMERG performance across varying temporal resolutions compared to ground rain gauge data.

2.3.1. Regression and Correlation Analysis

Regression analysis was used to model the linear relationship between the satellite estimates and gauge observations. Furthermore, the strength and direction of the linear relationship were quantified using the Pearson correlation coefficient (r). This metric evaluated how well the temporal variability of the satellite and gauge data matched. A high correlation showed that IMERG reliably captured the timing and pattern of rainfall events. Based on this perspective, the correlation coefficient was calculated as follows:

$$r = \frac{\sum_{i=1}^{n} (G_i - \overline{G})(S_i - \overline{S})}{\sqrt{\sum_{i=1}^{n} (G_i - \overline{G})^2 \sum_{i=1}^{n} (S_i - \overline{S})^2}} \quad (1)$$

where G_i is the gauge observation, S_i refers to the IMERG estimate, $\overline{G}$ and $\overline{S}$ denote respective means, and n shows the number of data pairs. A value of +1 implied a perfect positive linear relationship, while 0 suggested a nonlinear relationship [34].

2.3.2. Percent Bias (PBIAS)

The $PBIAS$ was calculated to measure the systematic tendency of satellite data. The essence was to overestimate or underestimate precipitation compared to the gauge measurements. This metric was crucial for water balance research, due to the quantification of long-term average bias in the dataset. With respect to the analysis, $PBIAS$ is defined as follows:

$$PBIAS = \frac{\sum_{i=1}^{n} (S_i - G_i)}{\sum_{i=1}^{n} G_i} \times 100\% \quad (2)$$

Following the description above, a $PBIAS$ value of 0% represents an unbiased estimate. Positive values show a systematic overestimation by IMERG, while negative values refer to an underestimation. The performance was generally considered excellent if $|PBIAS| < 10\%$, and satisfactory assuming $|PBIAS|$ was between 10% and 25% [34].

2.3.3. Nash–Sutcliffe Efficiency (NSE)

The NSE coefficient was adopted to further evaluate the predictive skill of the GPM IMERG precipitation estimates relative to ground-based observations. This widely applied performance metric in hydrological modeling measures how well the simulated or estimated values reproduce the observed variability relative to the mean of the observations. In addition, it is defined as stated in the following equation:

$$NSE = 1 - \frac{\sum_{i=1}^{n}(G_i - S_i)^2}{\sum_{i=1}^{n}(G_i - \overline{G})^2} \quad (3)$$

where G_i represents the observed rainfall from rain gauges, G_i is the corresponding IMERG estimate, $\overline{G}$ denotes the mean of the observed rainfall, and n refers to the number of paired observations.

The NSE value ranges from ∞ to 1.0, where a value of 1 shows a perfect match between simulated and observed data. Values between 0 and 1 represent acceptable model performance, while those less than 0 suggest that the mean of the observed data provides a better estimate than the satellite product. Following commonly accepted hydrological performance criteria [34], NSE values greater than 0.75 were considered excellent, between 0.65 and 0.75 were considered good, and between 0.5 and 0.65 were considered satisfactory, respectively. In addition, values less than 0.5 show unsatisfactory model performance.

2.3.4. Extreme Value Theory (EVT)

Block Maxima Analysis Using the Generalized Extreme Value (GEV) Distribution

An analysis based on EVT was conducted to evaluate the performance of IMERG in capturing extreme rainfall events. EVT provided a robust statistical framework for modeling the tail of the precipitation distribution and estimating the probability of rare, high-magnitude events [35]. Additionally, it enabled the characterization of extreme rainfall behavior by modeling the tail of the distribution, allowing reliable estimation of return levels for events whose recurrence periods exceeded the duration of the observed record [36]. The GEV distribution was fitted to the series of annual maximum daily rainfall values from the IMERG and rain gauge datasets. The cumulative distribution function (CDF) [37] of the GEV is stated as follows:

$$F(x;\mu,\sigma,\xi) = \exp\left\{-\left[1+\xi\left(\frac{x-\mu}{\sigma}\right)\right]^{-\frac{1}{\xi}}\right\}; \text{for } \left\{x : 1+\frac{\xi(x-\mu)}{\sigma} > 0\right\} \quad (4)$$

where μ is the location parameter, σ refers to the scale parameter (σ > 0), and ξ shows the shape parameter responsible for defining the behavior of the tail [38]. The parameters were estimated separately for the gauge and IMERG series using the L-moments or maximum likelihood method. Furthermore, the fitted distributions were used to calculate precipitation magnitudes for various return periods (e.g., 2, 10, 50, and 100 years). The return level XT for a T-year return period was derived by inverting the GEV distribution:

$$x_T = \mu - \frac{\sigma}{\xi}\left[1-\left(-\ln\left(1-\frac{1}{T}\right)\right)^{-\xi}\right], \text{for } \xi \neq 0 \quad (5)$$

The closeness of the return levels derived from IMERG to those from the gauge data exhibited the product's utility for extreme value analysis. This included flood frequency research and infrastructure design [39].

Peaks over Threshold (POT) Analysis

POT analysis was conducted to further strengthen the robustness of the extreme rainfall assessment and complement the block maxima method. The POT method used all independent exceedances above a predefined threshold, unlike the GEV method, which relied solely on annual maxima. The procedure increased the sample size of extreme events as well as improved statistical efficiency, particularly for brief or moderate-length datasets [38].

In this context, extreme rainfall events were identified by selecting all daily rainfall values exceeding a high threshold [40], based on a balance between statistical stability and sufficient sample size. The threshold was determined through exploratory analysis using the mean residual life plot (MRLP) and stability of parameter estimates, ensuring that exceedances represented genuine extremes while maintaining independence between events. A declustering procedure was applied to retain only the maximum value within each cluster of consecutive exceedances, further promoting independence.

The exceedances above the threshold were modeled using the Generalized Pareto Distribution (GPD). In addition, its CDF was expressed as follows:

$$F(x) = 1 - \left(1 + \frac{\xi(x-u)}{\sigma}\right)^{-1/\xi}, x > u \quad (6)$$

where u is the threshold, σ denotes the scale parameter ($\sigma > 0$), and ξ refers to the shape parameter governing the tail behavior [41]. Parameter estimation was performed using the maximum likelihood estimation (MLE) method. Furthermore, the GPD parameters were used to estimate return levels corresponding to selected return periods, allowing direct comparison with the GEV-based results.

The combined use of GEV and POT methods provided a more comprehensive characterization of extreme rainfall behavior. The POT method offered enhanced sensitivity to high-magnitude events and improved estimation reliability, particularly for longer return periods, while the GEV captured block-scale extremes. The agreement between the two methods increased confidence in the robustness of the extreme value estimates. The discrepancies outlined uncertainties associated with tail behavior and data limitations.

The integration of GEV and POT frameworks enabled this research to strengthen the assessment of extreme rainfall characteristics and improve the reliability of return-level estimates derived from IMERG and ground-based observations. Additionally, the combined use of block maxima (GEV) and threshold exceedance (POT) methods enabled a more robust evaluation of extreme rainfall behavior across different temporal and statistical scales.

3. Results and Discussion

3.1. Multi-Scale Evaluation of GPM IMERG Precipitation Performance

The performance of GPM IMERG precipitation estimates was evaluated against ground-based rain gauge observations across three watersheds. The evaluation process was conducted at both daily and monthly temporal scales using statistical indicators including the coefficient of determination (R^2), Pearson correlation coefficient (r), NSE, and PBIAS. The results of statistical performance across the research locations are shown in Table 2.

The results clearly showed that monthly aggregation substantially improved the agreement between GPM and gauge-based rainfall observations across all watersheds. Considering that performance at the daily scale remained moderate to low, the monthly-scale evaluation consistently exhibited high correlations and efficiency values, as shown in Figure 4. These were in line with previous results reported by [4,21,42–44]. The improvement outlined the capability of temporal aggregation to reduce random errors, spatial

mismatch, and timing inconsistencies inherent in satellite-based precipitation products. The monthly rainfall time series comparison in Figure 5 shows strong agreement between GPM IMERG and gauge observations, particularly in capturing seasonal variability and rainfall timing. GPM slightly underestimated some peak monthly values, enabling the overall consistency to remain high. This confirmed the suitability for monthly-scale hydrological applications. Additionally, discrepancies during extreme rainfall months outlined the inherent limitations of satellite retrievals in resolving localized convective events.

Table 2. Statistical performance metrics comparing IMERG precipitation estimates with ground-based rain gauge observations across the research location at daily and monthly temporal scales.

No	Watersheds	Catchment Area (km^2)	Rain Gauge Stations	Coef. Determination (R^2)		Correlation (r)		NSE	PBIAS
				Daily	Monthly	Daily	Monthly	Monthly	Daily and Monthly
1	Welang (Malang-Pasuruan)	511.6	10	0.32	0.92	0.57	0.96	0.94	−6.501
3	Kedak (Kediri)	52	3	0.23	0.91	0.48	0.96	0.87	15.91
4	Grindulu (Pacitan)	715	4	0.24	0.87	0.49	0.93	0.68	10.75

The Welang watershed (511.6 km^2, 10 rain gauges) had the best overall performance among the three research areas. At the daily scale, the coefficient of determination (R^2) = 0.323 and correlation coefficient (r) = 0.568 showed a moderate agreement between GPM and gauge observations. However, the performance improved substantially at the monthly scale, with R^2, r, and NSE = 0.924, 0.961, and 0.94, respectively, showing excellent model reliability. The PBIAS value of −6.50% suggested a slight underestimation by GPM, which remained within an acceptable range. The strong performance in this watershed was attributed to the relatively dense rain gauge network (10 stations) and moderate catchment size, which enhanced the representativeness of areal rainfall estimates as well as reduced spatial uncertainty.

The Kedak watershed (52 km^2, 3 rain gauges), characterized by small drainage area and sparse rain gauge distribution, showed weaker performance at the daily scale (R^2 = 0.227; r = 0.477). This reflected the high spatial variability of convective rainfall events that were difficult for satellite sensors to capture at the fine spatial scales. However, at the monthly scale, performance improved significantly, with R^2 = 0.914, r= 0.956, and NSE = 0.87, showing strong agreement. The positive PBIAS of 15.91% suggested systematic overestimation by GPM. This bias was associated with scale mismatch and the smoothing effect of satellite-based precipitation estimates over small catchments.

The Grindulu watershed (715 km^2, 4 rain gauges), despite being the largest among the three, exhibited moderate performance due to its relatively sparse rain gauge network. At the daily scale, R^2 and r values were 0.244 and 0.494, respectively, showing limited skill in capturing daily rainfall variability. The performance improved substantially (R^2 = 0.869; r= 0.932; NSE = 0.675) at the monthly scale, though not as strongly as in the Welang watershed. The positive PBIAS of 10.75% showed moderate overestimation, linked to spatial heterogeneity in rainfall patterns insufficiently represented by the limited number of gauges.

The results showed that rain gauge density exerted a stronger influence on GPM performance than the watershed area. In this context, watersheds with denser observation networks exhibited higher reliability in satellite–gauge agreement. The transition from daily to monthly temporal scales substantially enhanced statistical performance, reinforcing the suitability of GPM IMERG products for monthly hydrological and climatological analyses.

However, caution is required when applying GPM data at daily resolution, particularly for small catchments or extreme rainfall analyses. In such cases, bias correction or local calibration is strongly recommended to improve accuracy and reduce systematic errors. GPM IMERG is highly reliable for monthly-scale hydrological assessments, water balance research, and climate variability analysis. Rain gauge density played a more critical role than watershed size in determining the reliability of satellite-based precipitation estimates.

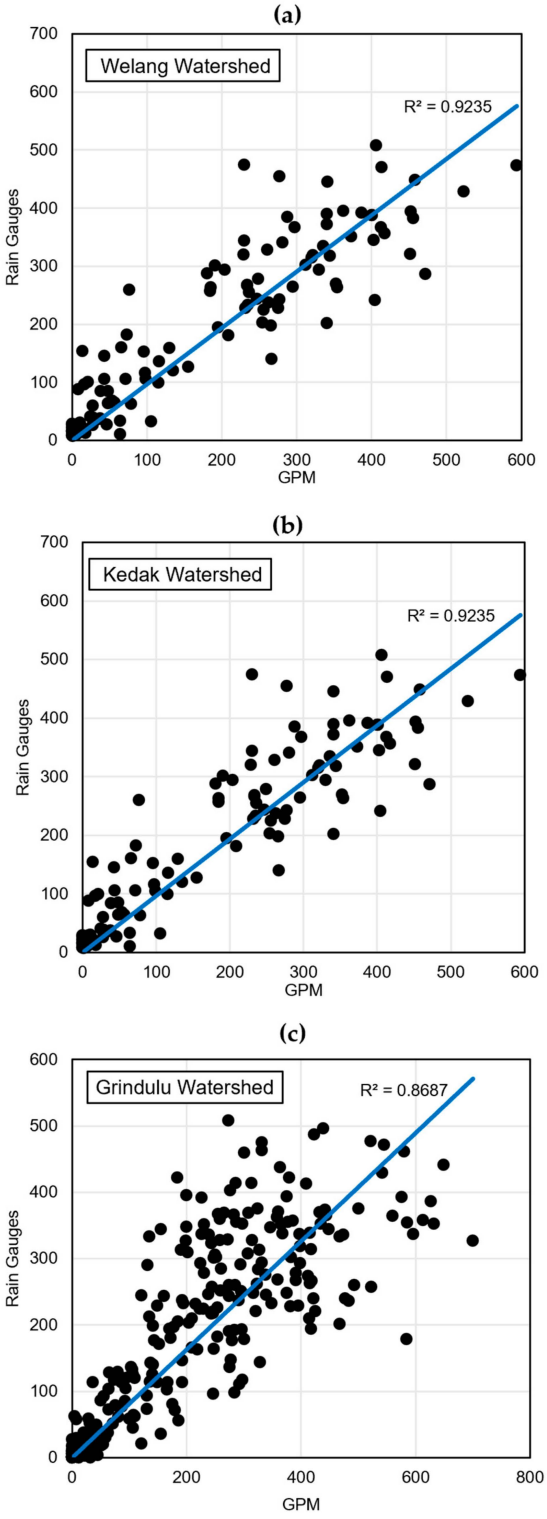


Figure 4. Monthly rainfall correlation of GPM IMERG and ground-based stations: (**a**) Welang, (**b**) Kedak, and (**c**) Grindulu watersheds.

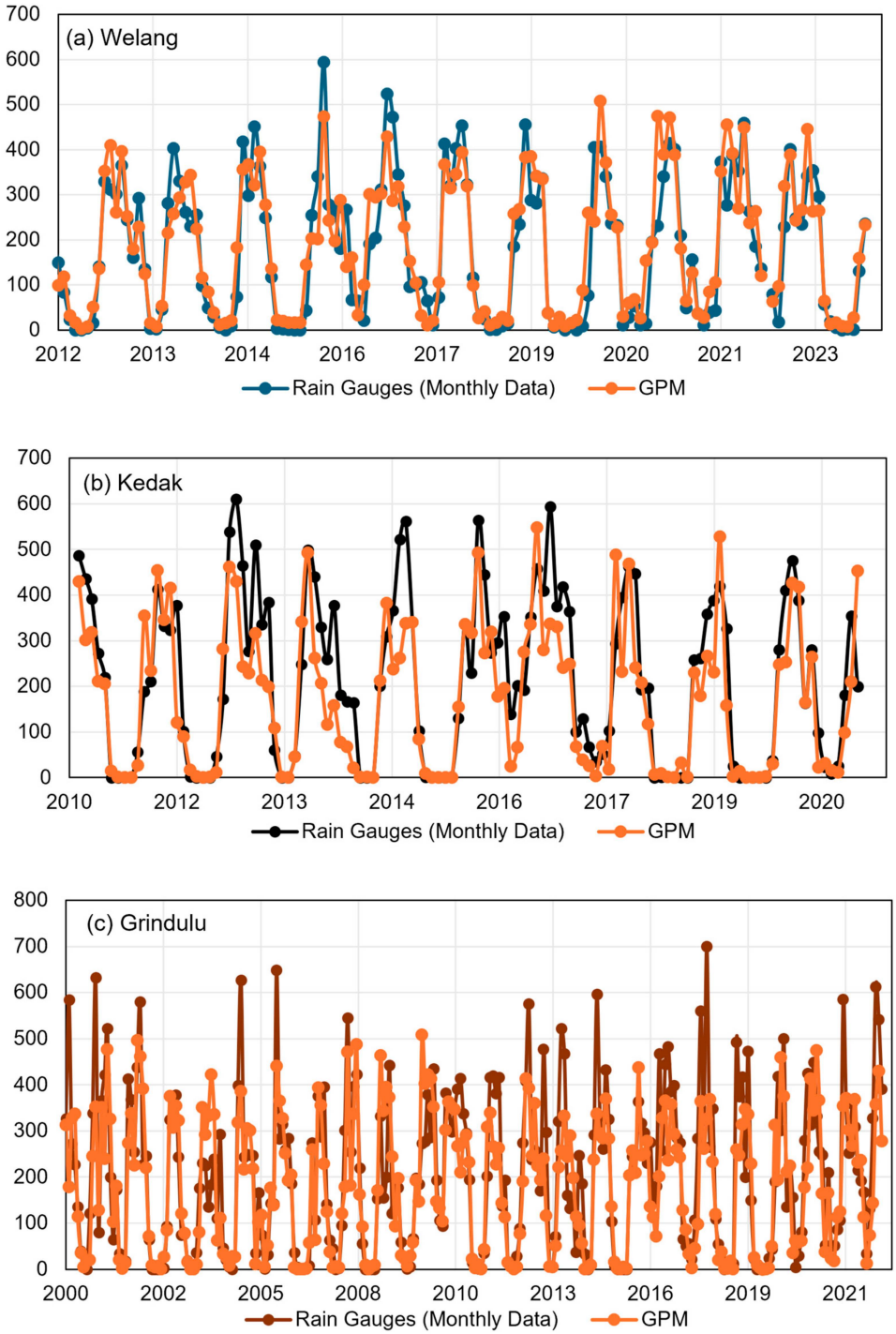


Figure 5. Monthly time series rainfall of GPM IMERG and ground-based stations: (**a**) Welang, (**b**) Kedak, and (**c**) Grindulu watersheds.

The CDF comparisons between GPM IMERG and ground-based rainfall observations across the Welang, Kedak, and Grindulu watersheds, as in Figure 6, showed a strong level of agreement. This suggested that GPM IMERG effectively reproduced the statistical structure of rainfall variability in the research region. The near-overlapping CDF curves, extremely high correlation coefficients (r = 0.998–0.999), and low RMSE values (<0.004) showed the strong consistency in the cumulative rainfall distributions derived from both datasets. This agreement was particularly pronounced in the mid-to-upper quantiles, which were critical for hydrological applications such as water balance estimation and reservoir operation. Minor deviations observed in the lower tail of the distribution reflected the

inherent limitations of satellite-based sensors in capturing low-intensity rainfall events. These deviations remained small and did not significantly affect cumulative rainfall volumes. The results showed that GPM IMERG was highly reliable for monthly and seasonal water balance assessments, reservoir management, and meteorological drought analysis. However, caution is warranted when applying the dataset to brief-duration or event-scale analyses, where local rainfall variability and temporal sequencing may not be fully resolved. The strong consistency between satellite-derived and ground-observed rainfall supported the robustness of GPM IMERG as a valuable data source for regional hydrological and water resource applications.

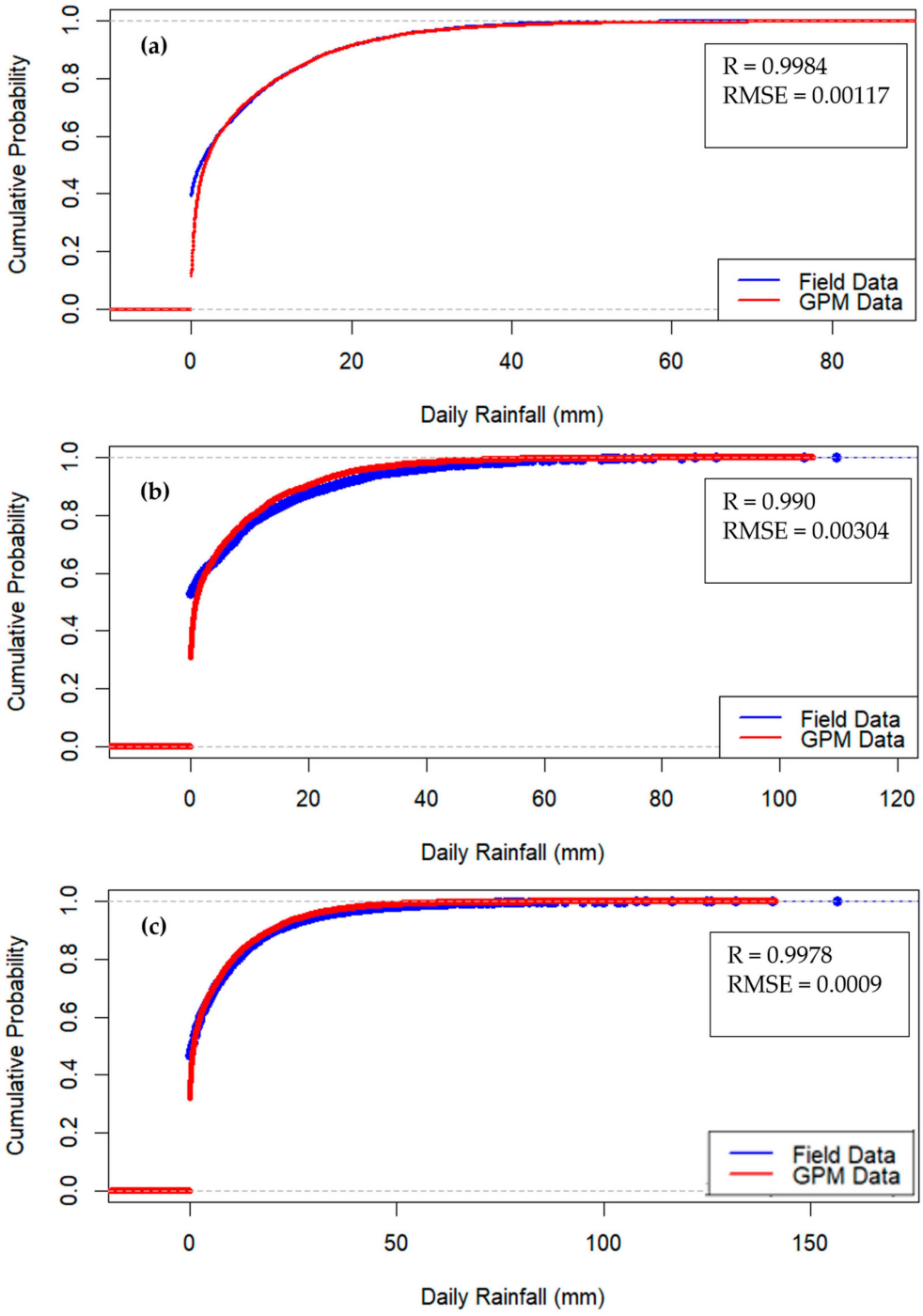


Figure 6. The CDF of GPM IMERG and ground-based rainfall: (**a**) Welang, (**b**) Kedak, and (**c**) Grindulu watersheds.

3.2. EVT Analysis

The EVT analysis in this study is evaluated using the Generalized Extreme Value (GEV) Annual Maxima (AM) and Peaks Over Threshold (POT) approaches. The GEV analysis revealed systematic and physically consistent differences between ground-based rainfall observations and GPM IMERG estimates across the Welang, Kedak, and Grindulu watersheds.

Based on the Smirnov–Kolmogorov test results shown in Table 3, both the ground-based rainfall observations and GPM IMERG data for the three watersheds (Kedak, Welang, and Grindulu) follow the Generalized Extreme Value (GEV) distribution. This is evidenced by the *p*-values for all six scenarios, which are all greater than the significance level of 0.05, and D-values below critical values. As noted, the GEV distribution fits the data for all cases, indicating that both ground and GPM data can be accurately modeled using the GEV distribution.

Table 3. Results of the GEV distribution GOF test using the Smirnov–Kolmogorov test.

No	Watershed	Data	*p*-Value	D-Value	Notes
1	Kedak	Field	0.7454	0.2007	The GEV distribution fits the data
		GPM	0.2561	0.3495	The GEV distribution fits the data
2	Welang	Field	0.1784	0.7787	The GEV distribution fits the data
		GPM	0.1714	0.8157	The GEV distribution fits the data
3	Grindulu	Field	0.1006	0.9559	The GEV distribution fits the data
		GPM	0.0904	0.9831	The GEV distribution fits the data

Despite variations in catchment size, gauge density, and data length, a common pattern emerged. GPM IMERG successfully reproduced interannual variability and moderate rainfall extremes, but it systematically underestimated high-magnitude events, particularly at longer return periods, as shown in Table 4. This behavior was primarily governed by differences in the tail characteristics of the fitted GEV distributions.

Table 4. Comparison of observed and GPM IMERG rainfall return periods and discrepancies across the Welang, Kedak, and Grindulu watersheds using GEV annual maxima.

No.	Return Period	Welang			Kedak			Grindulu		
		Rainfall Return Level (mm)		Discrepancies (%)	Rainfall Return Level (mm)		Discrepancies (%)	Rainfall Return Level (mm)		Discrepancies (%)
		Rain Gauges	GPM		Rain Gauges	GPM		Rain Gauges	GPM Data	
1	2	49.63	59.85	20.60	77.02	65.83	−14.52	94.11	71.70	−23.82
2	5	61.09	71.16	16.47	89.93	80.28	−10.73	118.34	88.54	−25.18
3	10	76.32	77.90	2.08	99.80	89.06	−10.76	132.10	99.51	−24.67
4	25	113.20	85.68	−24.31	113.99	99.33	−12.86	147.28	113.17	−23.16
5	50	163.84	90.94	−44.49	125.92	106.40	−15.51	157.12	123.15	−21.62
6	100	249.28	95.78	−61.58	139.11	112.98	−18.78	165.85	132.94	−19.84
Distribution		Frechet ($\xi > 0$)	Weibull ($\xi < 0$)		Frechet ($\xi > 0$)	Weibull ($\xi < 0$)		Weibull ($\xi < 0$)	Weibull ($\xi < 0$)	

In the Welang watershed (511.6 km^2, 12 years of data, and 10 rain gauges), the annual maximum rainfall derived from ground observations was generally lower than the estimated value from GPM data shown in Figure 7a. The return period analysis implied that the observed rainfall followed a Fréchet-type distribution ($\xi > 0$), reflecting heavy-tailed behavior and strong sensitivity to extreme rainfall events. However, GPM IMERG exhibited

a Weibull-type distribution ($\xi < 0$), as in Figure 7b, which resulted in a flatter return-level curve. The agreement between the two datasets was reasonable for short return periods (2–10 years). Divergence increased markedly for return periods exceeding 25 years, where GPM substantially underestimated extreme rainfall. The Fréchet behavior identified from the GEV analysis suggested potential instability in estimating long return periods beyond the data length, considering the relatively short observation period. This limitation outlined the need for complementary analysis using the POT method, which incorporated a larger number of extreme events, as well as provided a more robust characterization of extremes under data-scarce conditions.

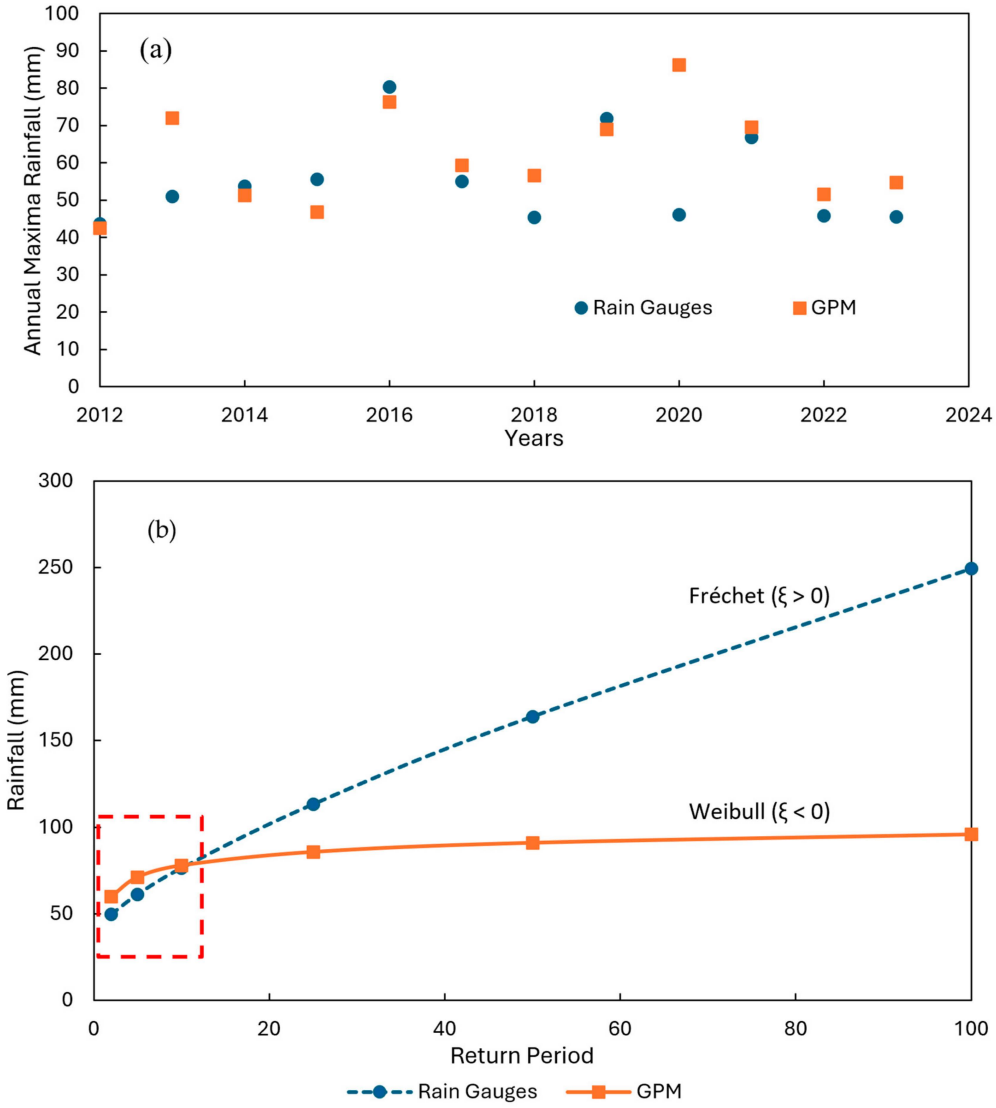


Figure 7. (**a**) Annual maxima of observed rainfall gauges and GPM data. (**b**) Rainfall return periods using GEV method (annual maxima data) in Welang watershed (red-dashed box indicates return period estimation are most stable within the 12-year observation range).

The structural discrepancy between the ground-based observations and the GPM IMERG satellite retrievals in the Welang and Kedak watersheds originates from a fundamental divergence in their respective statistical characterizations of extreme precipitation. While the field data exhibits a heavy-tailed Fréchet distribution, suggesting a vulnerability to high-magnitude stochastic extremes, the GPM IMERG data consistently follows a bounded Weibull distribution, which imposes a theoretical upper limit on rainfall intensity. This mismatch is primarily attributed to satellite retrieval constraints and the inherent

scales of tropical convective processes. Specifically, the GPM's 0.1° spatial resolution necessitates an area-averaging process that attenuates intense, localized convective cells across the pixel, effectively smoothing the peak intensities captured by point-source rain gauges. Furthermore, during extreme tropical events, the microwave sensors may reach signal saturation, and the prevalence of low-level maritime convection, which lacks the ice scattering required for robust satellite detection, can lead to a systematic underestimation of peak rainfall. Consequently, the satellite product generates a statistical signature that lacks the heavy-tail complexity necessary to reflect the true hydrological risk present at the ground level.

A similar but more pronounced pattern than in the Welang watershed was evident in the Kedak watershed (52 km^2, 10 years of data, and 3 rain gauges). Ground-based observations exhibited a Fréchet-type distribution ($\xi > 0$), showing strong convective control and high variability in extreme rainfall. Therefore, the GPM-derived distribution was bounded ($\xi < 0$), leading to an increasing underestimation of extreme rainfall with longer return periods. The discrepancy reached approximately 18–20% at the 100-year return period, as in Figure 8. Considering that the agreement between GPM and gauge observations remained reasonable for short return periods ($\leq$10 years), the performance deteriorated substantially for more extreme events. This behavior was mainly influenced by small catchment size, high spatial heterogeneity of rainfall, and limited gauge density, all of which clarified uncertainty in tail estimation. GPM adequately captured interannual variability; however, its applications in extreme-event estimation associated with small catchments must be approached with caution, especially for long return periods.

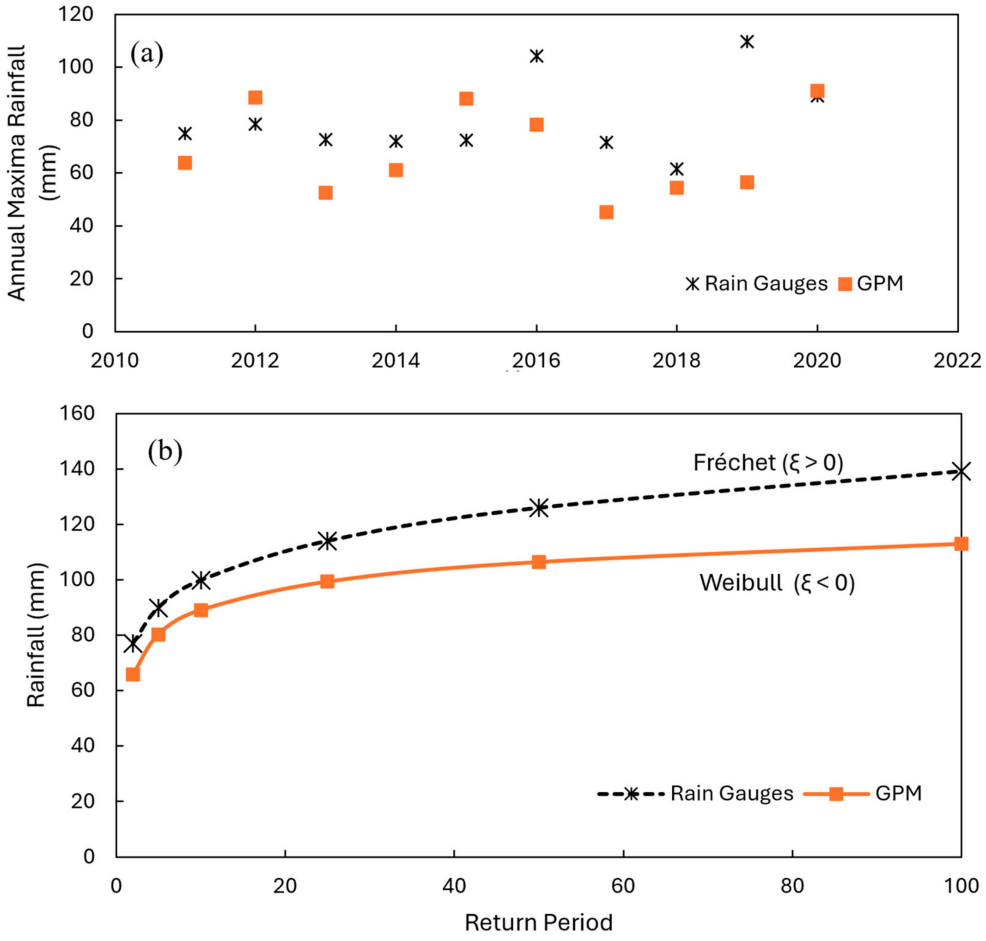


Figure 8. (**a**) Annual maxima of observed rainfall gauges and GPM data. (**b**) Rainfall return periods using GEV method (annual maxima data) in Kedak watershed.

Both field and GPM datasets in the Grindulu watershed (715 km^2, 20 years of data, and 4 rain gauges), exhibited Weibull-type distributions, representing bounded extremes, as in Figure 9. Although GPM continued to underestimate high return levels, the magnitude of bias was reduced (approximately 20–25%) compared to the smaller basins. The larger watershed area and longer observation period contributed to more stable EVT behavior and improved consistency between satellite and ground-based estimates. However, the smoothing effect inherent to satellite retrievals remained evident, even at these spatial scales.

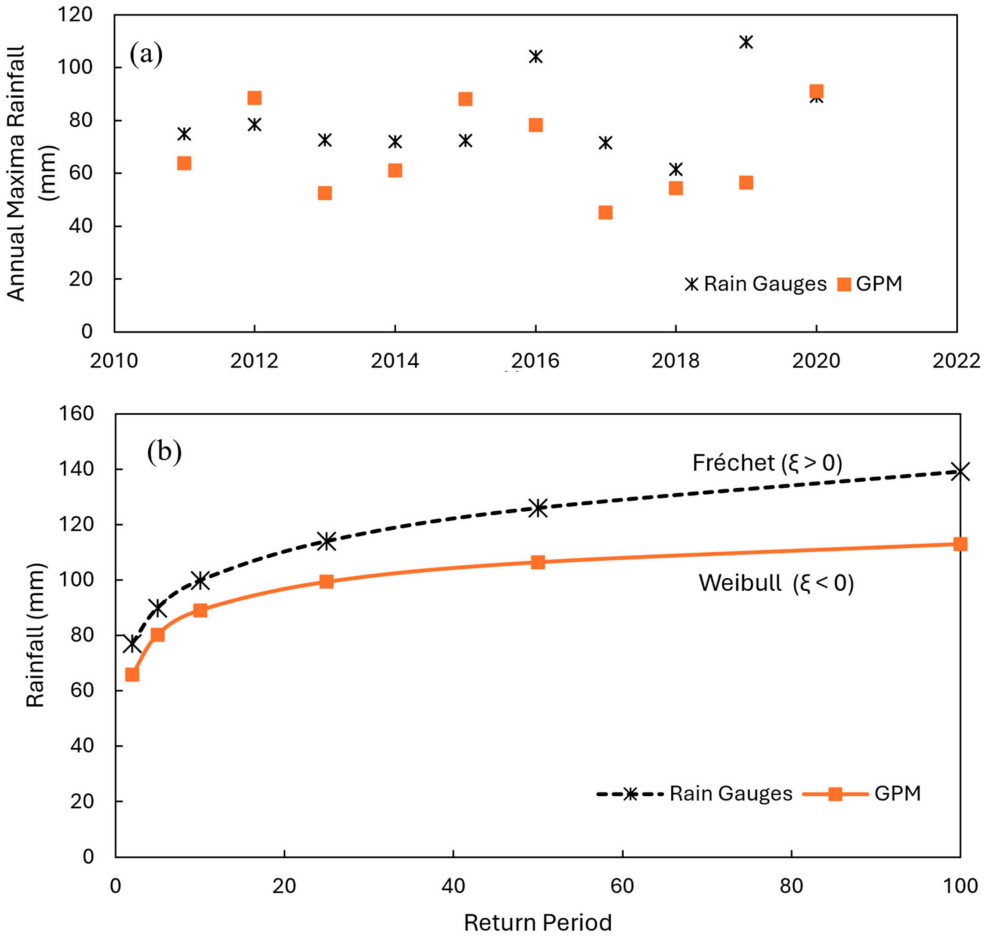


Figure 9. (**a**) Annual maxima of observed rainfall gauges and GPM data. (**b**) Rainfall return periods using GEV method (annual maxima data) in Grindulu watershed.

These results showed that the reliability of GPM IMERG for extreme rainfall estimation was strongly controlled by watershed scale, data availability, density of observed rain gauge stations, and statistical nature of rainfall extremes. GPM systematically underestimated rare extremes, particularly in small and convectively dominated basins, despite the adequate performance for moderate events and regional-scale assessments. The applications of satellite-based precipitation products for hydrological design and risk assessment should incorporate ground-based calibration or bias-correction strategies. The essence was to ensure robust and physically meaningful estimates of extreme rainfall.

The AMS-based EVT framework provided valuable insight into the behavior of annual extremes, despite being inherently limited by its reliance on a single maximum value per year. This method may underutilize available information, particularly in regions characterized by frequent high-intensity rainfall events. However, a POT analysis was applied to address the limitations and further examine the robustness of extreme rainfall characterization. The POT method enabled the inclusion of multiple exceedances above

a physically meaningful threshold, offering enhanced sensitivity to tail behavior and improved representation of extreme rainfall processes. This complementary analysis allowed for a more comprehensive evaluation of the consistency between ground observations and GPM IMERG estimates across varying hydrological regimes.

The POT analysis provided a refined characterization of extreme rainfall behavior by explicitly modeling exceedances above a selected threshold. This method enabled the efficient use of available data and yielded more stable estimates, particularly under limited record length, compared to the AMS–GEV framework. The results showed systematic and physically consistent differences between ground-based rainfall observations and GPM IMERG estimates across the three watersheds (Welang, Kedak, and Grindulu). GPM reproduced moderate extremes and exhibited persistent biases at higher return periods. In addition, the magnitude and direction varied by watershed scale and rainfall regime.

Threshold selection and statistical diagnostics confirmed the robustness of the POT framework applied. Based on this perspective, thresholds were selected within ranges exhibiting near-linear behavior in the MRLP (Figure 10). Candidate values were systematically evaluated using the GPD. The optimal threshold for each watershed was determined in accordance with a combination of criteria (Table 5). This included the Mann–Kendall test *p*-values greater than 0.05, minimum Akaike Information Criterion (AIC) values, and satisfactory goodness-of-fit statistics ($p > 0.05$). The resulting thresholds, ranging from approximately 25 to 35 mm, which were produced between 185 and 430 exceedance events, offered sufficient sample sizes for stable and reliable parameter estimation. Kolmogorov–Smirnov (KS) test values exceeding 0.80 across all cases suggested strong agreement between empirical and modeled distributions. Moreover, Mann–Kendall statistics showed stronger monotonic behavior in gauge-based observations (0.57–0.60) compared to GPM estimates (0.14–0.25), reflecting weaker temporal persistence in satellite-derived extremes. Lower AIC values for gauge data further confirm superior model performance relative to GPM, particularly in larger basins. These diagnostics collectively showed that the observed discrepancies between gauge and satellite estimates arose from physical and structural differences in rainfall representation rather than deficiencies in statistical fitting.

Table 5. Summary of POT model parameters and goodness-of-fit statistics for observed and GPM IMERG rainfall across the watersheds.

No	Watershed	Threshold (mm)		Number of Exceedances	Mann Kendall_*p*-Value	AIC	Kolmogorov–Smirnov_*p*-Value
1	Welang	Observed Rain Gauges	25.26	227	0.60	1472.82	0.82
		GPM	25.51	223	0.25	1519.94	0.86
2	Kedak	Observed Rain Gauges	34.24	185	0.60	1369.88	0.98
		GPM	26.93	185	0.22	1299.79	0.99
3	Grindulu	Observed Rain Gauges	31.50	413	0.57	3225.64	1.00
		GPM	27.90	430	0.14	3069.77	0.98

Return period comparisons derived from POT, as in Table 6, showed distinct watershed-dependent behaviors. In the Welang watershed (511.6 km^2), gauge-based return levels increased from 57.8 mm (2-year) to 85.9 mm (100-year). Meanwhile, GPM estimates rose from 65.3 mm to 108.4 mm, representing systematic overestimation by values ranging from 13% to 27%. This contradicted the AMS-based results, where GPM tended to underestimate extremes. In the Welang watershed, the contrast between the POT- and GEV-based results further outlined the influence of sample size and spatial aggregation on extreme rainfall estimation. Considering that the GEV analysis relied on only 12 annual

maxima, the POT framework incorporated substantially more information, with 277 and 233 exceedances for gauge observations and GPM, respectively. This larger effective sample size improved statistical stability and reduced uncertainty in tail estimation. GPM consistently overestimated return levels relative to gauge observations under the POT framework. This contradicted the GEV results, where GPM tended to underestimate extremes. The shift suggested that the limited length of the annual maxima series in the GEV analysis clarified uncertainty at long return periods, particularly in the presence of heavy-tailed behavior. In the Welang basin, the presence of 10 rain gauges provided relatively dense spatial coverage, allowing area-averaged observations to effectively capture localized convective extremes. Consequently, the observed rainfall field exhibited lower spatial smoothing than the satellite-derived estimates, which inherently averaged rainfall over larger footprints. Regarding the POT framework, gauge-based extremes were lower than GPM estimates despite the contradictory behavior observed under GEV. The results proved that in data-rich basins such as Welang, POT offered a more stable and physically consistent representation of extremes, while discrepancies with satellite products reflected structural differences in spatial sampling rather than purely statistical artifacts. The results reinforced the importance of combining EVT methods with an understanding of measurement scale and spatial representativeness when interpreting extreme rainfall behavior.

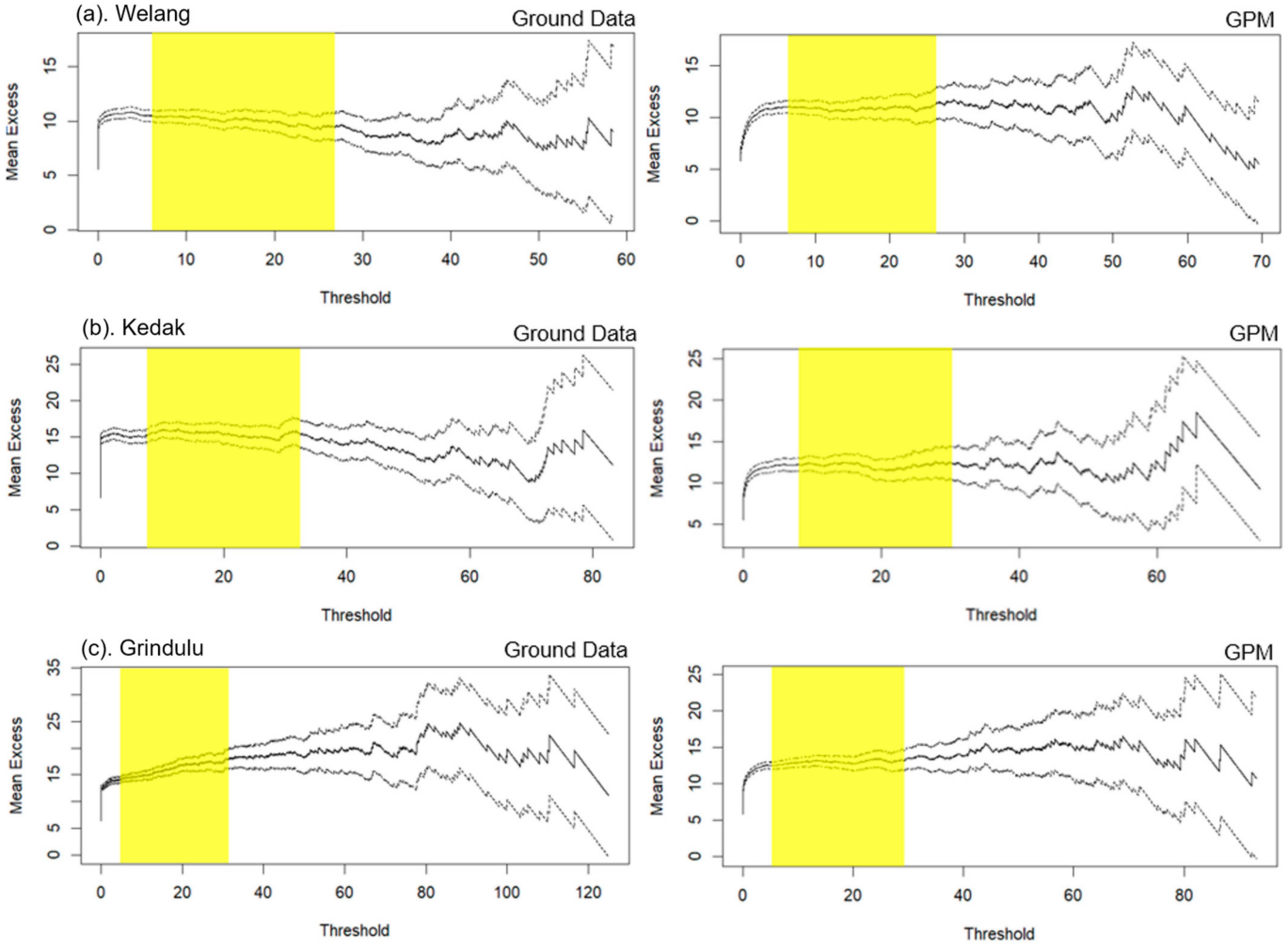


Figure 10. Mean Residual Life Plots (MRLP) used for threshold selection in the Peaks Over Threshold (POT) analysis for the (**a**) Welang, (**b**) Kedak, and (**c**) Grindulu watersheds.

Table 6. Comparison of rainfall return periods derived from POT analysis using rain gauge observations and GPM IMERG data across the watersheds.

No.	Return Period	Welang			Kedak			Grindulu		
		Rainfall Return Level (mm)		Discrepancies (%)	Rainfall Return Level (mm)		Discrepancies (%)	Rainfall Return Level (mm)		Discrepancies (%)
		Rain Gauges	GPM		Rain Gauges	GPM		Rain Gauges	GPM	
1	2	57.75	65.33	13.12	83.25	70.62	−15.17	98.63	77.90	−21.02
2	5	64.94	75.42	16.14	92.84	81.43	−12.29	117.32	92.71	−20.97
3	10	70.13	83.05	18.42	99.46	89.53	−9.99	131.89	104.58	−20.71
4	25	76.69	93.14	21.45	107.47	100.14	−6.82	151.76	121.19	−20.14
5	50	81.43	100.78	23.75	113.00	108.09	−4.34	167.24	134.48	−19.59
6	100	85.99	108.41	26.07	118.12	115.99	−1.81	183.15	148.45	−18.94

In the Kedak watershed (52 km^2), the POT results showed slight discrepancies between gauge and GPM estimates compared to Welang. GPM underestimated rainfall by approximately 2–15% across return periods, with the least bias occurring at longer return periods, as in Table 6. This relatively improved agreement reflected lower absolute rainfall magnitudes and reduced variability in extreme events. The brief observation record (10 years) and low Mann–Kendall statistic (0.22) suggested limited temporal persistence, showing that uncertainty remained substantial. As a result, the POT framework improved consistency relative to the AMS method, prompting the reliability of long-term extrapolation in this small catchment to remain constrained.

The Grindulu watershed (715 km^2) exhibited the most stable and consistent behavior among the three research sites. Gauge-based return levels increased from 98.6 mm (2-year) to 183.2 mm (100-year). GPM-derived values were within the range of 77.9 mm and 148.5 mm, in line with the systematic underestimation of approximately 19–21%, as in Table 6. The large number of exceedances (>400) and the lengthy observation period substantially enhanced statistical robustness, as reflected by high KS *p*-values (>0.98) and consistent tail behavior between datasets. Although GPM still underestimated extremes, the magnitude of bias was significantly reduced compared to smaller basins. This outlined the stabilizing effect of spatial averaging and increased data availability. The results proved that basin scale and data length played a critical role in controlling the reliability of satellite-based extreme rainfall estimates. Additionally, the comparison of rainfall return period values with observed rain gauge and GPM data using the POT analysis is shown in Figure 11.

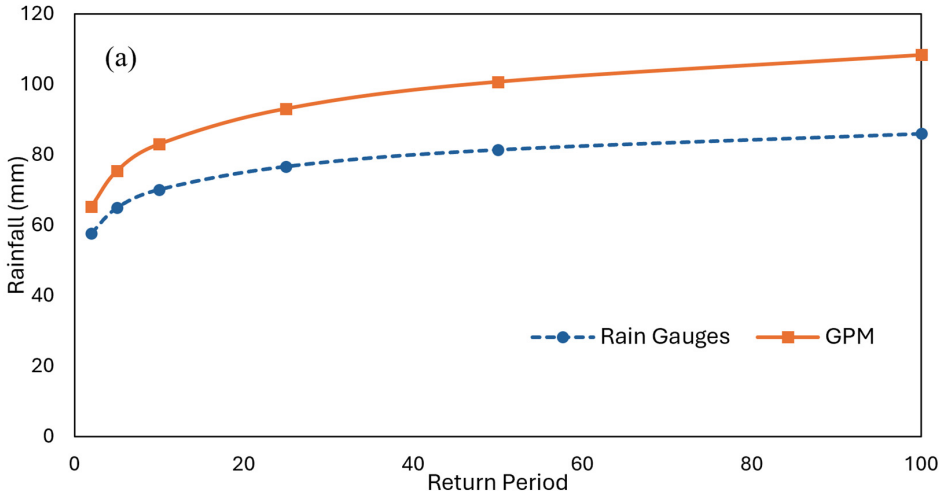


Figure 11. *Cont.*

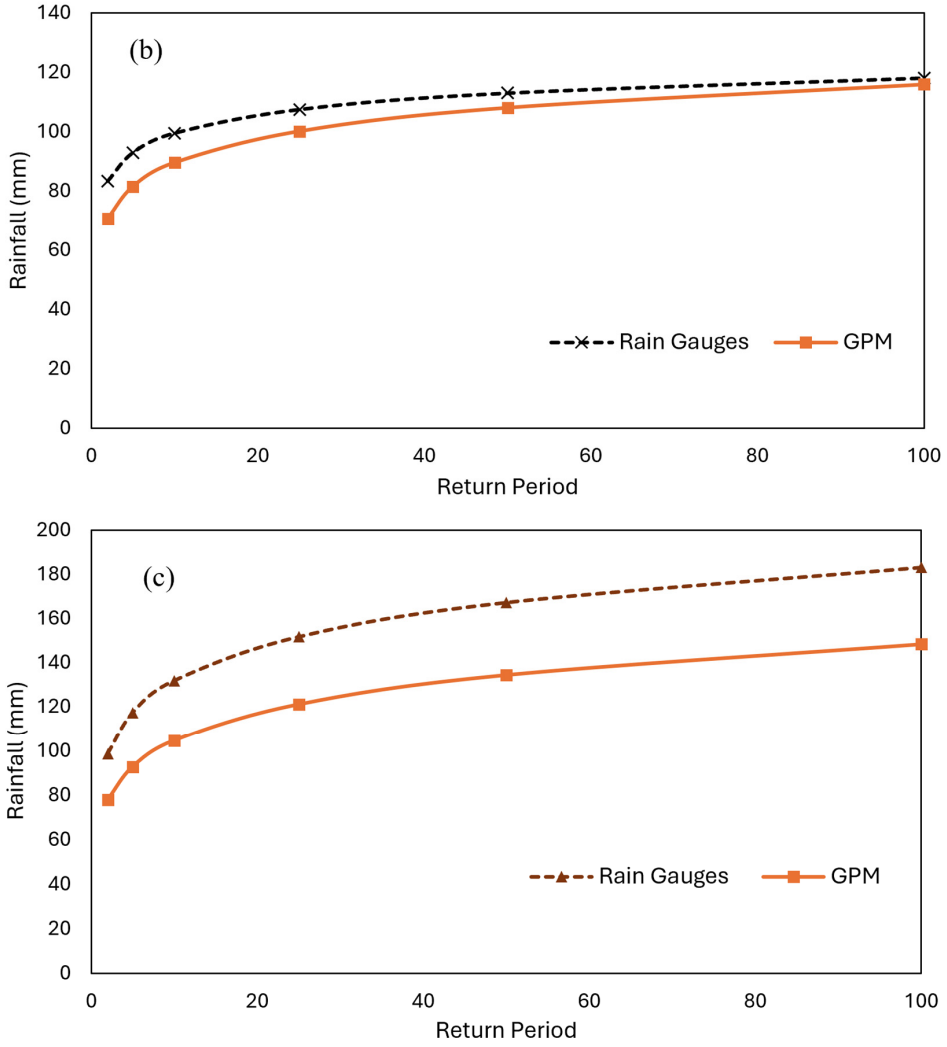


Figure 11. Rainfall return periods using POT method: (**a**) Welang, (**b**) Kedak, and (**c**) Grindulu watersheds.

The comparison between POT and GEV frameworks showed that POT provided smoother and more stable extreme value estimates, particularly for short- or moderate-length records. GPM IMERG reliably captured moderate rainfall extremes (2–10-year return periods), as well as systematically underrepresented higher-order extremes, specifically in small and convectively dominated catchments. Although satellite-based precipitation products were suitable for regional-scale hydrological assessment, the direct applications for design-level analysis (≥25-year return periods) required caution. Bias correction, gauge–satellite merging, or hybrid EVT frameworks played essential roles in ensuring robust and physically consistent extreme rainfall estimation for flood risk mitigation and management in Indonesia.

3.3. Implications for Sustainable Flood Risk Management

The findings of this EVT analysis have profound, practical implications for flood risk management and infrastructure design in tropical catchments. Standard hydrological design codes for critical infrastructure, such as dam spillways, river flood analysis, and other water resource infrastructures, rely on accurate estimations of high-return-period extreme rainfall (e.g., 50-year or 100-year events).

Because GPM IMERG inherently smooths localized convective peaks, resulting in a bounded Weibull tail rather than the heavy-tailed Fréchet distribution observed at ground stations using uncorrected, raw satellite data, it systematically and dangerously underes-

timates design rainfall depths. For local stakeholders and water resource engineers, the actionable recommendations are as follows:

a. Application of Tail-Specific Bias Correction
   Raw GPM IMERG data must not be used directly for extreme hydrological design in small-to-medium East Java catchments without applying a tail-specific bias correction factor or an explicit engineering safety margin.
b. Methodological Selection for EVT
   When analyzing satellite data for flood design, the POT method should be prioritized over the GEV (block maxima) approach for larger catchments (like Welang), as POT better captures the spatial extent of frequent, high-intensity events that GEV might discard.
c. Early Warning System Limitations
   Due to the severe underestimation of localized daily extremes, uncorrected GPM IMERG should not be used as the sole trigger mechanism for flash-flood Early Warning Systems (EWS) in fast-responding, steep catchments like Kedak.

The strong correlation between GPM and ground rainfall observations at the monthly scale can serve as a reliable reference for monthly water resource analysis and management. This includes practical applications such as water balance analysis for irrigation allocation, the determination of cropping patterns, monthly reservoir water balance, and the estimation of dependable discharge. Consequently, these findings provide a valuable framework for stakeholders in Indonesia to support future sustainable water management using satellite precipitation data

4. Conclusions

In conclusion, this research provided a comprehensive assessment of GPM IMERG precipitation performance across multiple watersheds in East Java using statistical evaluation and EVT. The results showed that IMERG performed robustly at monthly temporal scales, with correlation coefficients exceeding 0.90, NSE values reaching 0.94, and low bias ($|PBIAS| < 10\%$) in well-instrumented catchments. The close agreement in CDF ($R \approx 0.99$; $RMSE < 0.004$) confirmed the capability of IMERG to reproduce long-term rainfall variability, supporting its applicability for regional-scale hydrological and climatological assessments such as water balance and dependable discharge analysis.

At the daily scale, performance deteriorated substantially, particularly in small and convectively dominated watersheds. The correlation values were less than 0.60. This depicted the limited ability of satellite-based retrievals to resolve localized high-intensity precipitation.

The results of the GEV analysis demonstrated that the observed rainfall followed a heavy-tailed Fréchet-type distribution ($\xi > 0$) in the Welang and Kedak watersheds, based on 12 and 10 years of rainfall data, respectively. Consequently, this resulted in a significant increase in values for return periods that exceed the length of the data record (return periods over 10 years). Caution is required here, as this significant upward trend renders rainfall predictions for extended return periods less accurate when relying on a GEV model with a data length of only 10 to 12 years. Meanwhile, a Weibull-type behavior ($\xi < 0$) was identified in the Grindulu watershed. GPM exhibited a bounded Weibull-type behavior ($\xi < 0$), which led to the systematic underestimation of extreme rainfall. This underestimation became pronounced at longer return periods, depending on watershed scale. Based on the relatively brief observation record, the Fréchet behavior inferred from the gauge data suggested potential instability in estimating return levels beyond the available data length. GPM substantially underestimated extreme rainfall, with part of the divergence at long return periods reflecting uncertainty associated with extrapolating extremes from limited

observations. These results showed the need to be cautious when interpreting long-period return levels, including the importance of complementary methods, such as POT analysis, to improve the robustness of extreme rainfall estimation.

The POT-based analysis showed that extreme rainfall characterization was strongly controlled by watershed scale, data availability, and spatial representativeness of observations. GPM IMERG consistently exhibited systematic biases relative to gauge observations, with the magnitude and direction of these biases varying by catchment size and rainfall regime across the three watersheds. In the Welang watershed, the larger number of exceedances and dense gauge network yielded more stable estimates, proving GPM tended to overestimate extremes under the POT framework due to spatial averaging effects that smooth localized convective rainfall. However, the smaller Kedak watershed had reduced discrepancies and higher uncertainty, reflecting limited data length and weaker temporal persistence. The Grindulu watershed exhibited the strongest behavior, with lengthy records and large spatial extent enhancing statistical stability. This led to reduced bias, alongside GPM underestimating high return levels. The results collectively confirmed that POT provided a more reliable and physically consistent representation of extreme rainfall than AMS, particularly when supported by sufficient exceedance samples. The remaining discrepancies between gauge- and satellite-based estimates outlined the importance of accounting for spatial resolution, sampling density, and basin scale when applying satellite precipitation products for extreme rainfall and hydrological risk assessments.

The results demonstrate that GPM IMERG is suitable for monthly-scale hydrological analyses and regional climate applications, though it remains inadequate for direct use in design-level extreme rainfall estimation. Robust engineering practice requires the implementation of bias correction, gauge–satellite merging, or hybrid Extreme Value Theory (EVT) frameworks to ensure accuracy. These findings underscore the necessity of integrating ground observations with satellite data to improve the characterization of extreme precipitation in tropical environments. Consequently, this study establishes a methodological framework for stakeholders in Indonesia to integrate enhanced satellite products into resilient water management and sustainable infrastructure design.

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Abbreviations

The following abbreviations are used in this manuscript:

AIC	Akaike information criterion
AM	Annual maxima
AMS	Annual maximum series
CDF	Cumulative distribution function
CMORPH	Climate Prediction Center morphing method
DMC	Double mass curve
EVT	Extreme value theory
EWS	Early warning systems
GEV	Generalized extreme value
GPCC	Global Precipitation Climatology Centre
GPD	Generalized Pareto distribution
GPM	Global precipitation measurement
IMERG	Integrated multi-satellite retrievals
JAXA	Japan Aerospace Exploration Agency
KS	Kolmogorov–Smirnov
MLE	Maximum likelihood estimation
RMSE	Root mean square error
MRLP	Mean residual life plot
NASA	National aeronautics and space administration
NSE	Nash–Sutcliffe efficiency
PBIAS	Percent bias
POT	Peaks over threshold
QC	Quality control
TMPA	TRMM multi-satellite precipitation analysis
TRMM	Tropical rainfall measuring mission.

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